

Problem Set 5

1

No one likes a cold seat! The smart toilet's engineers designed a resistive heater embedded in the seat, which converts current into heat using Joule heating (like a toaster). However, the inductance of this structure is wreaking havoc on the power supply, making it so users must choose between using heat or the odor extraction fan.

a

The magnetic field of the given structure is predicted by:

$$\vec{\mathbf{B}} = \hat{z} \frac{2\mu_0}{\pi D} \left(\ln \left(\frac{8D}{d} \right) - 2 \right) I$$

This model assumes $\mathbf{B}$ is constant over the whole region (it's not) and that there is no magnetic field in the resistive wires (not true), but it's good enough for toilet problems.

For $N = 2$ turns, $D = 30$ cm, $d = 3$ mm, and $I = 10$ A, predict the heater coil inductance.

✓ Answer ✓

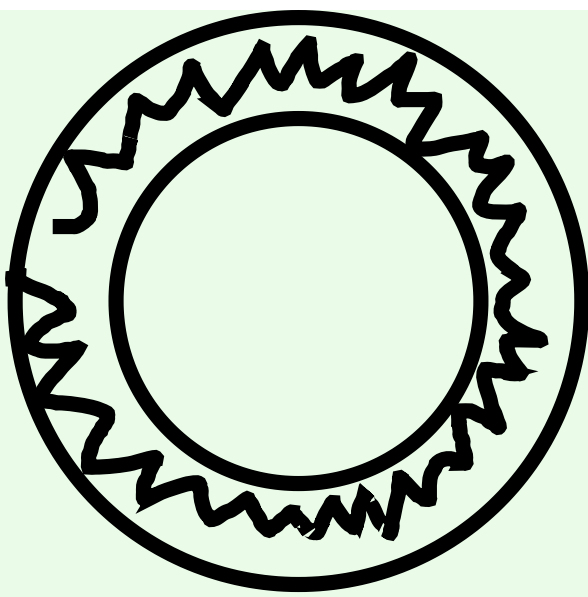
$$\begin{aligned} L &= \frac{N(\vec{B} \cdot \vec{A})}{I} \\ B \cdot A &= \frac{8\pi \cdot 10^{-7}}{\pi \cdot 0.3} \left(\ln \left(\frac{240}{0.3} \right) - 2 \right) 10 (\pi \cdot 0.15^2) \\ &= \frac{8 \cdot 10^{-7}}{0.3} (\ln(800) - 2) 10 (\pi \cdot 0.15^2) \\ L &= 1.766057014627458 \text{ mH} \end{aligned}$$

b

To remove the inductance issue, sketch out a different routing path for the coil which would use about the same amount of resistive wire, but with much lower inductance. Hint: draw a flux surface and use the right hand rule to estimate the net flux in all directions.

Note: The center region of the seat has to stay free of wire for obvious reasons.

✓ Answer



As the alternating directions of the wire would cancel out the magnetic fields

2

The switch in the bottom loop of Fig. P6.1 is closed at $t = 0$ and then opened at a later time t_1 . What is the direction of the current I in the top loop (clockwise or counterclockwise) at each of these two times?

✓ Answer

If we are treating the two wires nearby's nearby effects then the current in the top loop will be the same rotation direction as the bottom one (clockwise or counterclockwise).

If we are treating both wires as an inductor coil as a whole, then the rotation directions will be opposite.

I believe this question wants us to treat for nearby effects, so when the switch is closed then both loops go clockwise, and when the switch opens, both would go counterclockwise.

3

A circular-loop TV antenna with 0.02 m^2 area is in the presence of a uniform-amplitude 300 MHz signal. When oriented for maximum response, the loop develops an emf with a peak value of 30 (mV). What is the peak magnitude of B of the incident wave?

✓ Answer

$$\text{emf}(t) = -B_0 A \omega \cos(\omega t)$$

$$B_0 = \frac{30 \text{ mV}}{A \omega} = 795.7747154594768 \text{ pT}$$

4

The electromagnetic generator shown in Fig. 6-12 is connected to an electric bulb with a resistance of $150\ \Omega$. If the loop area is $0.1\ \text{m}^2$ and it rotates at 3600 revolutions per minute in a uniform magnetic flux density $B_0 = 0.4\ \text{T}$, determine the amplitude of the current generated in the light bulb.

✓ **Answer**

$$3600\ \text{rpm} = 60\ \text{Hz}$$

$$s(t) = B_0 A \sin(\omega t)$$

$$\text{emf}(t) = -B_0 A \omega \cos(\omega t)$$

$$V = 0.4(0.1)(2\pi(60)) = 15.079644737231009\ \text{V}$$

$$I = \frac{V}{R} = 0.10053096491487339\ \text{A}$$